IMO Training – Number Theory
Primes and Theorems on Congruences

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Notes

Unless otherwise specified, let p be a prime number.

Theorem. (Fermat's Little Theorem) $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$. Equivalently, $\forall a \in \mathbb{Z}$ with $(a,p) = 1$, $a^{p-1} \equiv 1 \pmod{p}$.

Definition. (Euler Totient Function) $\forall n \in \mathbb{N}$, $\varphi(n) = \#\{m \in \mathbb{N} \mid 1 \leq m < n, (m,n) = 1\}$. Equivalently, for $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ being distinct primes, and integers $a_1, a_2, \dots, a_k \geq 1$, $\varphi(n) = n\left(1 - \dfrac{1}{p_1}\right)\left(1 - \dfrac{1}{p_2}\right) \dots \left(1 - \dfrac{1}{p_k}\right)$.

Theorem. (Fermat-Euler Theorem) $\forall a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $(a,n) = 1$, $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem. (Wilson's Theorem) p is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

Example. Let p be a prime number, and $n \in \mathbb{Z}$. If $p \mid n^2$, prove that $p^2 \mid n^2$.

Example. Let $p > 5$ be a prime number. Prove that $240 \mid p^2 - 1$.

Example. Find all prime numbers p s.t $p+2, p+4$ are both prime numbers.

Example. Find all $n \in \mathbb{N}$ s.t $3n-4, 4n-5, 5n-3$ are all prime numbers.

Example. Find the remainder when 2^{1000} is divided by 13.

Example. Find the remainder when 5^{100} is divided by 221.

Example. Find the last two digits of 7^{1234}.

Example. Find all prime numbers p such that $11 \mid 2^p + 3^p$.

Example. Prove that there does not exist a prime number p such that $p \mid 2^{p-1} + 3^{p-1}$.

Example. Find all primes p, q such that $pq \mid p^q + q^p + 7$.

Example. Find all $m, n \in \mathbb{Z}$ s.t. $m^3 - n^3 = 2013$.

Exercises

1. (IMO 1989 Problem 5) Prove that $\forall n \in \mathbb{N}$, there exists n consecutive positive integers none of which is an integral power of a prime power.

2. Prove that $\varphi(2^n - 1)$ is divisible by n.

3. Find a positve integer x s.t. $x^{13} = 21982145917308330487013369$.

4. Let $p \equiv 3 \pmod 4$ be a prime. Prove that $a^2 + b^2 \equiv 0 \pmod p$ implies $a \equiv b \equiv 0 \pmod p$.

5. Find all pairs of primes p, q s.t. $pq \mid (5^p - 2^p)(5^q - 2^q)$.

Extra Space